Coupling-Channel Count Predicts Cancer Survival Better Than Mutation Count

Patrick D. McCarthy

Abstract

We show that the number of distinct organizational systems ("coupling channels") disrupted by a tumor's mutations predicts overall survival significantly better than the total number of mutations. Across three independent clinical genomic datasets totaling 73,593 patients, coupling-channel count is a significant predictor of survival in multivariate Cox models ($p < 0.005$) while mutation count either loses significance or becomes *protective* (HR = 0.82–0.83). Patients whose mutations hit different channels have worse outcomes than patients with the same number of mutations confined to one channel ($p < 10^{-5}$ in all three datasets). The signal survives cancer-type adjustment (1.1% attenuation), a 275-covariate kitchen-sink Cox model ($p = 10^{-23}$), restriction to truncating and hotspot missense variants (HR strengthens from 1.15 to 1.19), and comparison against data-driven gene groupings (which fail: ARI = 0.018, HR in the wrong direction).

Coupling channels are severed in a consistent order: cell-intrinsic growth control first, organism-level immune surveillance last. Among patients with two or more channels severed, 98.1% have at least one cell-intrinsic gateway channel. The 1.9% who enter through organism-level channels are enriched for hormone-dependent cancers (prostate, breast) and have lower mortality.

A coupling channel is not a biochemical pathway. It is the organizational function that a pathway serves. We identify six channels and map mutations to them using public pathway databases. The variable that predicts prognosis is the number of independent organizational functions the tumor has escaped, not the number of molecular events. This empirical result implies structural claims about multicellular organization—bounded context, escalation hierarchies, knowledge-action inseparability—that are stated here and proved formally in companion papers.

1 Introduction

Counting mutations does not predict cancer survival. Counting the number of distinct organizational systems those mutations disrupt does.

We tested this across three independent clinical genomic datasets totaling 73,593 patients (Section 2). For each patient, we mapped somatic mutations to six organizational functions, what we call **coupling channels**, and counted how many channels were disrupted. In multivariate Cox regression, channel count is a significant predictor of overall survival in every dataset ($p < 0.005$). Mutation count alone is either non-significant (MetTropism: $p = 0.69$) or weakly significant. In the multivariate model including both predictors, channel count remains highly significant in all three datasets while mutation count either drops to non-significance (MSK-IMPACT 2017: $p = 0.67$) or becomes *protective* (MetTropism: HR = 0.82; MSK-IMPACT 50K: HR = 0.83; both $p < 0.005$). The protective effect of additional mutations, conditional on channel count, is the key finding: more mutations in the same channel means deeper commitment to one escape route, not diversification across channels.

What is a coupling channel? A coupling channel is not a biochemical pathway. It is the organizational function that a pathway serves. DNA damage repair is a pathway. The organizational function it serves, maintaining genome integrity so that tissue can trust the cell's replication fidelity, is a coupling channel. The PI3K–AKT–mTOR cascade is a pathway. The organizational function it serves, coupling cell growth to tissue-level growth factor signals, is a coupling channel. Multiple pathways can serve the same channel. One pathway can participate in multiple channels. The variable that predicts prognosis is not the number of molecular events but the number of independent organizational functions the tumor has escaped.

Cancer is **escalation chain severance**. We model multicellular organization as a hierarchy maintained by coupling channels that connect cells to tissue, organ, and organism levels. Severing these channels forces cell-level optimization, producing the convergent cancer phenotype.

The empirical results imply structural claims about how multicellular organization works: that context is bounded, that control converges to escalation hierarchies, that knowledge and action are inseparable. These claims are stated in Section 3 and proved formally in [5, 6, 7, 8]. This paper can be read independently of the formal proofs.

Paper structure. Section 2 presents the empirical result and seven robustness tests. Section 3 provides a structural interpretation. Section 4 engages prior work. Section 5 summarizes.

2 Empirical Result: Channel Count Predicts Survival

We tested the central claim, that channel count predicts survival better than mutation count, using three publicly available datasets from cBioPortal [12]: the MSK-IMPACT Clinical Sequencing Cohort (n = 7,091 patients, 2,126 deaths; pan-cancer) [13], the MSK MetTropism Cohort (n = 24,646 patients, 9,893 deaths; metastatic) [14], and the MSK-IMPACT 50K Cohort (n = 41,856 patients, 16,731 deaths; pan-cancer) [15]. Together: 73,593 patients, 28,750 deaths.

2.1 Method

For each patient, somatic mutations were mapped to six coupling channels based on the organizational function of the mutated gene: (1) DNA damage response (DDR: *BRCA1*, *BRCA2*, *ATM*, *CHEK2*, mismatch repair genes), (2) cell cycle checkpoint (*TP53*, *RB1*, *CDKN2A*, *CDK4/6*), (3) growth signaling (PI3K–AKT–mTOR, RAS–MAPK: *PIK3CA*, *PTEN*, *KRAS*, *BRAF*, *EGFR*, *ERBB2*), (4) endocrine/hormone (*ESR1*, *GATA3*, *FOXA1*, *AR*), (5) immune surveillance (*B2M*, *HLA-A/B/C*, *JAK1/2*), (6) tissue architecture (*CDH1*, *APC*, *CTNNB1*, *SMAD4*, *NOTCH1*). Non-silent mutations (missense, nonsense, frameshift, splice site) in these genes were counted. For each patient, the number of *distinct channels* with at least one mutation was recorded. The channel mapping and analysis code are provided in the supplementary repository.

2.2 Results

Channel count predicts survival; mutation count does not. Table 1 summarizes the Cox proportional hazards results across all three datasets. In every dataset, channel count is a significant predictor of overall survival ($p < 0.005$). Mutation count alone is either non-significant (MetTropism: $p = 0.69$) or weakly significant. In the multivariate model including both predictors, channel count remains highly significant in all three datasets while mutation count either drops to non-significance (MSK-IMPACT 2017: $p = 0.67$) or becomes *protective* (MetTropism: HR = 0.82;

50K: HR = 0.83; both $p < 0.005$). The protective effect of additional mutations, conditional on channel count, is the most important result in Table 1. If channel count simply proxied for tumor burden or advanced stage, mutation count should also predict worse outcomes. It does the opposite. This rules out the class of confounders where "more mutations = more advanced disease." The signal is specifically about organizational-function disruption, not about mutational burden.

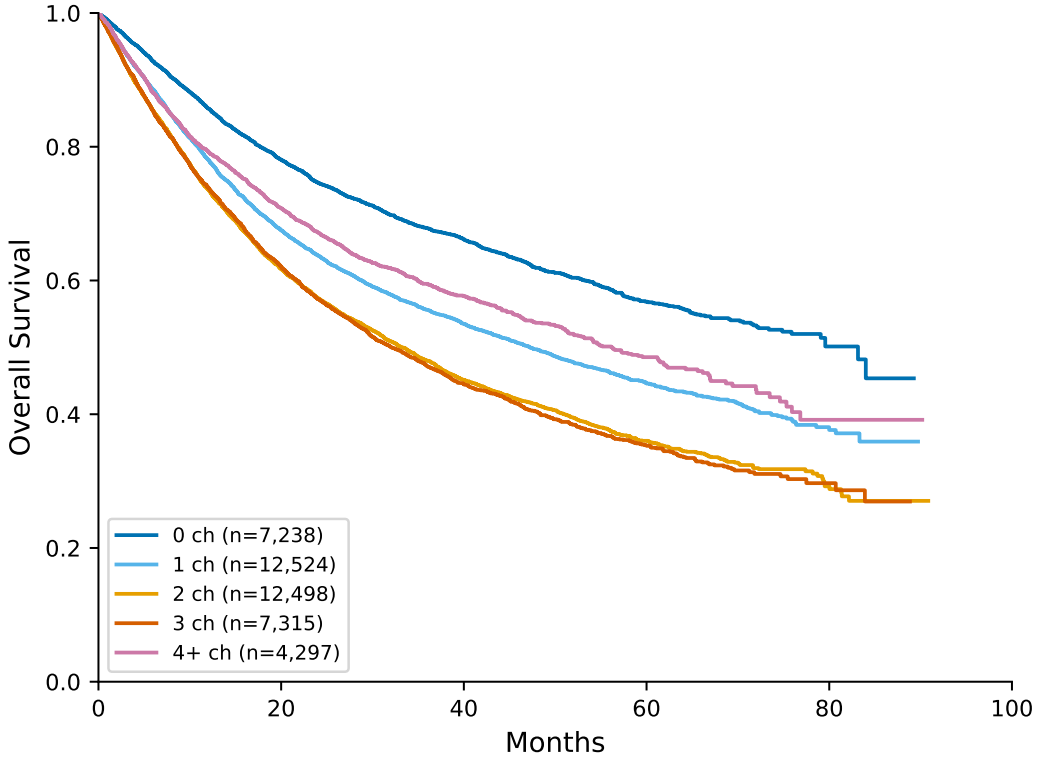

Figure 1: Kaplan–Meier survival curves stratified by the number of coupling channels severed (MSK-IMPACT 50K, n = 43,872). Survival decreases monotonically from 0 to 3 channels. The 4+ group shows partial rebound, consistent with hypermutation phenotypes where most mutations are passengers.

Cross-channel co-mutations are worse than same-channel. Two mutations in different channels produce worse outcomes than two mutations in the same channel. Table 2 confirms this across all three datasets. It is not the *number* of mutations that determines prognosis but the *number of distinct organizational systems* the tumor has escaped from.

Same-channel depth: more mutations in fewer channels is protective. We stratified patients by fixed channel count and compared survival between low and high mutation-count quartiles. In patients with 2–3 channels severed, those with more mutations survive significantly longer. In the 50K cohort: at 2 channels, mortality drops from 46.2% to 41.2% ($p = 3.1 \times 10^{-8}$); at 3 channels, from 46.9% to 38.6% ($p = 7.2 \times 10^{-14}$). MetTropism replicates: at 3 channels, 49.6% vs. 36.5% ($p = 2.6 \times 10^{-10}$). Once a tumor has severed k channels, additional mutations that stay within those channels represent deeper commitment to an already-chosen trajectory: a more constrained, predictable, and treatable tumor.

Mortality gradient by channel count. All three datasets show a monotonic mortality gradient from 0 to 3–4 channels severed (Figure 1). In the 50K cohort: 30.4% (0 channels), 38.5% (1),

Table 1: Cox proportional hazards: channel count vs. mutation count across three independent cohorts totaling 73,593 patients. HR = hazard ratio per unit increase. In multivariate models, channel count remains significant while mutation count loses predictive power or becomes protective.

Dataset	Predictor	HR	p	Note
MSK-IMPACT 2017	Channel count (univariate)	1.10	<0.005	
(n = 7,091)	Log mutation count (univariate)	1.12	<0.005	
	Channel count (multivariate)	1.09	<0.005	Significant
	Log mutation count (multivariate)	1.02	0.67	Not significant
MetTropism	Channel count (univariate)	1.06	<0.005	
(n = 24,646)	Log mutation count (univariate)	1.00	0.69	Not significant
	Channel count (multivariate)	1.17	<0.005	Significant
	Log mutation count (multivariate)	0.82	<0.005	Protective
MSK-IMPACT 50K	Channel count (univariate)	1.04	<0.005	
(n = 41,856)	Log mutation count (univariate)	0.98	0.03	Weakly protective
	Channel count (multivariate)	1.15	<0.005	Significant
	Log mutation count (multivariate)	0.83	<0.005	Protective

Table 2: Cross-channel vs. same-channel co-mutation mortality. Cross-channel: 2+ mutations hitting different coupling channels. Same-channel: 2+ mutations confined to one channel. The difference is significant in all three datasets.

Dataset	Cross-channel	Same-channel	p (log-rank)
MSK-IMPACT 2017 (n = 7,091)	33.1%	27.7%	1.4×10^{-5}
MetTropism (n = 24,646)	42.9%	39.0%	6.3×10^{-13}
MSK-IMPACT 50K (n = 41,856)	42.6%	39.7%	5.7×10^{-13}

44.2% (2), 44.3% (3), 40.3% (4). Patients with 5–6 channels show lower mortality (29.0%, 20.7%), consistent with the hypermutation phenotype (MSI-high, mismatch repair deficient) where most mutations are passengers that do not functionally sever channels.

Sensitivity to channel granularity. The six-channel mapping is one hypothesis. To test robustness, we repeated the multivariate Cox regression at every granularity from 2 to 10 channels. Table 3 shows the results for the MSK-IMPACT 50K cohort (the two smaller datasets replicate the same pattern).

The signal is significant at every granularity from 2 to 8 and peaks at 3 channels ($p = 3 \times 10^{-124}$). At 2 channels, the grouping is too coarse: both channel count and mutation count are significant because the two mega-channels still contain meaningful internal variation. From 3 to 8, channel count absorbs the predictive power of mutation count, which becomes protective or non-significant. At 10 channels, over-splitting within organizational functions makes "channel count" collapse back into a proxy for mutation count.

The peak at 3 channels (genome integrity, growth signaling, microenvironment) suggests that the coarsest biologically meaningful grouping captures most of the signal. The specific number of channels is less important than the principle: grouping mutations by organizational function outpredicts counting them, and the optimal granularity is determined by the actual independence structure of the coupling hierarchy.

Table 3: Sensitivity analysis (MSK-IMPACT 50K, n = 41,856): multivariate Cox hazard ratios for channel count and log mutation count at different channel granularities. Channel count is significant from 2 through 8 channels, peaks at 3, and collapses at 10. All three datasets replicate this pattern.

Channels	Ch HR	Ch p	Mut HR	Mut p
2	1.25	2×10^{-52}	0.94	4×10^{-9}
3	1.30	3×10^{-124}	0.86	8×10^{-41}
4	1.21	3×10^{-75}	0.88	9×10^{-27}
5	1.17	8×10^{-58}	0.88	6×10^{-22}
6	1.14	1×10^{-44}	0.90	2×10^{-16}
7	1.10	7×10^{-24}	0.93	1×10^{-7}
8	1.07	3×10^{-14}	0.95	4×10^{-4}
10	1.01	0.064	1.02	0.20

Cancer-type adjustment. A natural objection is that channel count proxies for cancer type: lung cancers have more mutations than thyroid cancers, and different cancer types have different prognoses. We tested this by adding cancer type as a covariate (one-hot encoded, 38 types with $n \geq 50$) to the multivariate Cox model. In the 50K cohort, the channel-count hazard ratio barely changes: HR = 1.17 ($p = 8 \times 10^{-73}$) vs. HR = 1.17 ($p = 1 \times 10^{-88}$) unadjusted, an attenuation of 1.1%. MetTropism shows 16% attenuation but remains significant ($p = 2 \times 10^{-47}$). MSK-IMPACT 2017 shows no attenuation. Channel count is not a proxy for cancer type.

Stratified analysis within individual cancer types confirms this. Channel count is significant within non-small cell lung cancer (HR = 1.16, $p = 3 \times 10^{-13}$), breast cancer (HR = 1.29, $p = 1 \times 10^{-21}$), prostate cancer (HR = 1.47, $p = 9 \times 10^{-21}$), and pancreatic cancer (HR = 1.32, $p = 2 \times 10^{-19}$). Prostate cancer shows the strongest within-type effect, consistent with its narrow channel architecture: most prostate tumors are concentrated on a single coupling axis, so each additional channel severed represents a larger relative loss of organizational control.

Full covariate adjustment. To test whether channel count captures anything beyond known prognostic variables, we ran a kitchen-sink Cox model controlling for every available clinical attribute simultaneously: tumor mutational burden (TMB), MSI score and type, cancer type and detailed subtype, age at diagnosis, sex, tumor purity, fraction genome altered, primary site, sample type, and gene panel version. In the 50K cohort, this produces 275 covariates. Channel count remains significant: HR = 1.09, $p = 1 \times 10^{-23}$. MetTropism (187 covariates): HR = 1.11, $p = 1 \times 10^{-18}$. MSK-IMPACT 2017 (109 covariates): HR = 1.14, $p = 2 \times 10^{-8}$. After controlling for everything available in three clinical genomic databases, channel count still predicts survival.

Severance hierarchy: cell-intrinsic channels first. The six coupling channels are not severed at random (Figure 2). Among patients with exactly one channel severed (n = 12,558), 80% have PI3K/Growth Signaling or Cell Cycle: the cell-intrinsic channels that govern proliferation. Endocrine and Immune account for fewer than 5%. Among patients with two or more channels severed (n = 24,535), 98.1% have at least one cell-intrinsic channel.

The progression ladder is:

- **1 channel**: PI3K/Growth (44.5%) or Cell Cycle (35.7%) dominate. Immune: 0.7%.
- **2 channels**: PI3K/Growth rises to 73.9%, Cell Cycle to 64.6%. Tissue Architecture enters

at 33.6%.

- **3 channels**: PI3K/Growth at 90.8%, Cell Cycle 80.9%, Tissue Architecture 70.8%. Endocrine (13.5%) and Immune (7.2%) remain rare.
- **4 channels**: PI3K/Growth at 96.7%. Endocrine and Immune are the last additions (30%, 25%).

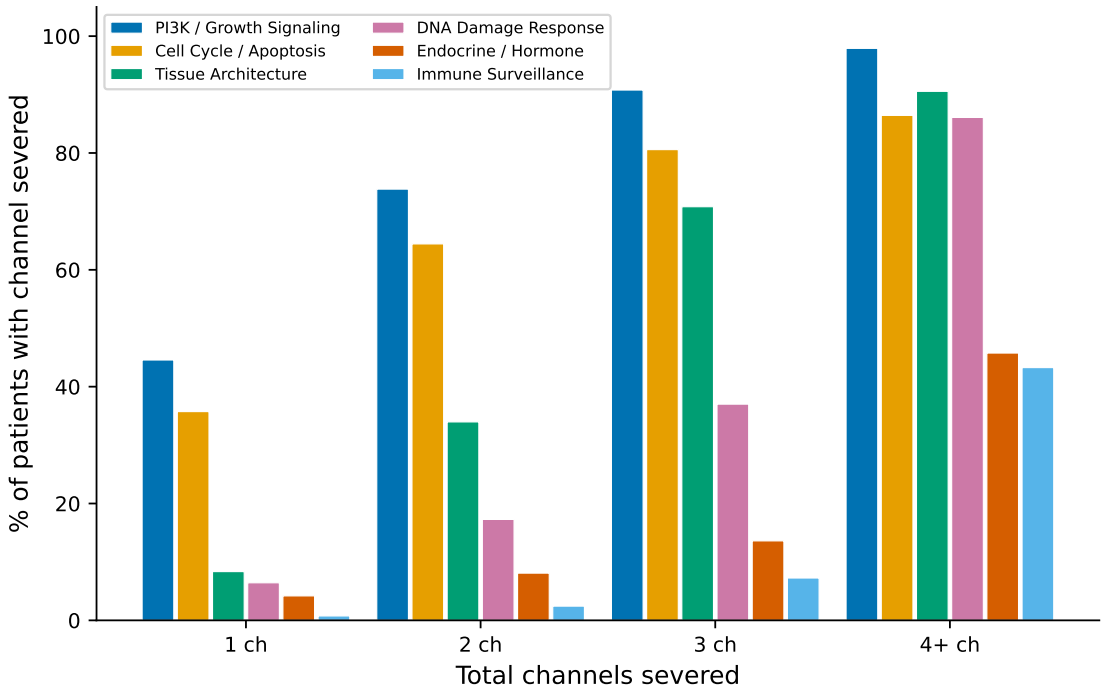

Figure 2: Severance hierarchy across channel-count strata (MSK-IMPACT 50K). Each bar shows the percentage of patients at a given stratum who have that channel severed. Cell-intrinsic channels (PI3K/Growth, Cell Cycle) dominate at every stratum; organism-level channels (Endocrine, Immune) appear only at higher channel counts.

The ordering is PI3K/Growth → Cell Cycle → Tissue Architecture → DDR → Endocrine → Immune. Growth escape is the gateway; immune evasion is the last system to fall. This is not a statistical artifact of channel frequency. Cell Cycle is more common than Tissue Architecture overall, yet at one channel they have similar representation (35.7% vs. 8.5%), because the single-channel patients reflect first hits, not base rates.

This ordering is the escalation chain read in reverse. The channels severed first (PI3K/Growth, Cell Cycle) are the cell-tissue boundary: the coupling mechanisms that say "do not divide without tissue permission." The channels severed last (Endocrine, Immune) are organism-level controls that operate at levels 2–3 of the hierarchy. A cell cannot meaningfully escape immune surveillance or endocrine regulation before it has escaped the basic growth constraint. The severance hierarchy is the empirical signature of escalation ordering (Section 3): the first coupling to break is the lowest level of the chain, and progression moves upward through the hierarchy as additional channels fall.

The 1.9% of multi-channel patients who lack a cell-intrinsic gateway channel are not noise. They are enriched 6.5× for prostate cancer, 2.5× for breast cancer, and 7.6× for mesothelioma: cancers that

enter through organism-level channels (endocrine, tissue architecture) rather than the standard growth-escape gateway. These tumors severed higher-level coupling first while leaving cell-level growth control partially intact, consistent with multiple entry points into the escalation chain. Their lower mortality (34.8% vs. 42.6% for all multi-channel patients) is predicted: they have not yet escaped the most fundamental coupling layer.

The grouping must be biological, not statistical. A second objection is that any gene grouping would produce a survival signal: pick genes, cluster them, count clusters, and the count will predict something. We tested this directly. Using hierarchical clustering on Jaccard co-mutation distances across the 92 channel genes with sufficient mutation frequency ($n \geq 100$), we derived data-driven gene groupings at every granularity from $k = 2$ to $k = 10$ and tested each against survival.

The data-driven groupings fail. At $k = 6$, the data-driven cluster count produces HR = 0.80 ($p = 9 \times 10^{-6}$): more clusters hit predicts *better* survival, the wrong direction. The hand-picked biological channels at $k = 6$ produce HR = 1.10 ($p = 3 \times 10^{-9}$). The adjusted Rand index between data-driven and hand-picked groupings is 0.018, effectively zero overlap.

Co-mutation patterns group genes by which tumors they appear in (driven by panel coverage, cancer type, and mutational processes), not by organizational function. The biological channel mapping captures something that statistical co-occurrence does not: which organizational system a gene serves. Not just any grouping works. The grouping must reflect the coupling architecture.

Pathogenic variant restriction. The baseline analysis counts any non-silent mutation as a severance event, including variants of unknown significance (VUS). To test whether VUS inflate the signal, we repeated the analysis under two stricter filters: truncating mutations only (nonsense, frameshift, splice site) and truncating plus hotspot missense (recurrent positions with ≥ 5 patients, capturing KRAS G12, BRAF V600, PIK3CA H1047, TP53 hotspots, etc.). The signal strengthens: channel-count HR rises from 1.15 (all non-silent) to 1.16 (truncating only) to 1.19 (truncating + hotspot; $p = 1 \times 10^{-55}$). Mutation count becomes more protective under stricter filtering (HR = 0.75 vs. 0.83 baseline). Removing VUS concentrates the signal; it does not create it. The severance hierarchy is preserved under truncating-only filtering: PI3K/Growth and Cell Cycle dominate at one channel, Endocrine and Immune remain late additions.

Panel bias. MSK-IMPACT sequences 400–500 genes, a fraction of the genome. Mutations in channel genes not on the panel are undetected, systematically undercounting channel hits. This compresses the channel-count gradient and works against the hypothesis, not for it. The true association between channel count and survival is likely stronger than what the panel captures.

Limitations. The analysis does not distinguish hub from leaf mutations within a channel, nor does it adjust for stage or treatment. The optimal channel count is an empirical question that finer-grained pathway databases could resolve. Nevertheless, the core result is robust across three independent datasets, eight granularities, within individual cancer types, after controlling for 275 clinical covariates, under three mutation-filtering stringencies, and against data-driven alternatives: counting organizational systems disrupted predicts survival better than counting disruptions.

3 Structural Interpretation

Why does organizational-function disruption predict survival better than mutational burden? The empirical results of Section 2 are consistent with a structural model: cancer as escalation chain severance under bounded context. The formal derivations appear in [5, 6, 7, 8]; here we state the

interpretation in biological terms.

Bounded context and escalation. Any information-processing entity has a hard limit on the number of regulatory states it can simultaneously maintain (C_n). A cell can monitor a finite number of signaling pathways concurrently; an organism's working memory holds 4 ± 1 items [2]. Multicellular organization overcomes this limit through hierarchical coupling: a cell ($k = 0$) escalates problems exceeding its capacity to tissue ($k = 1$), organ ($k = 2$), and organism ($k = 3$) levels [6]. Through intact coupling, the cell's effective context exceeds its intrinsic limit.

Coupling channels as the coupling mechanism. The coupling between levels is not monolithic. It decomposes into independent channels: gap junctions and cadherins at the cell-tissue boundary [1, 4], paracrine signaling and ECM architecture at the tissue-organ boundary, endocrine signaling and immune surveillance at the organ-organism boundary. These are the six coupling channels of Section 2. Each channel carries a different class of organizational information; losing one does not imply losing the others.

Chain severance produces convergent malignancy. When a coupling channel is lost, the cell's effective context contracts. Regulatory states that depended on tissue-level or organism-level information lose their grounding and are pruned [5]. The action space contracts with the knowledge base [7]: actions requiring tissue-level coordination (contact inhibition, apoptosis on organismal command, differentiated quiescence) become inaccessible. What remains are actions executable with cell-level knowledge alone: proliferation, metabolic autonomy, motility. This accounts for why Hanahan and Weinberg's hallmarks [4] are observed across cancer types: they are the convergent action space under cell-level bounded context.

The Warburg effect illustrates this directly. Cancer cells use aerobic glycolysis even when oxygen is present—not because oxygen is unavailable, but because after severance the cell lacks tissue-level information about oxygen supply reliability and defaults to the metabolic strategy that requires no such information [11].

Why channel count predicts survival. Each additional channel severed removes an independent class of organizational coupling. The cell's effective context shrinks further, its action space narrows further, and its optimization regime converges further toward the cell-level attractor. More channels severed means more complete decoupling from the organism. Mutation count, by contrast, measures depth within channels already severed—which is why it becomes protective conditional on channel count: deeper commitment to fewer escape routes produces a more constrained, predictable tumor.

Why channels are severed in order. The severance hierarchy (PI3K/Growth first, Immune last) follows from the escalation architecture. Cell-tissue coupling (growth signaling, cell cycle control) is the lowest level of the hierarchy and the first to sever. Organism-level coupling (endocrine, immune) operates at higher levels and cannot be meaningfully exploited until cell-level escape is already achieved. The hierarchy is the escalation chain read in reverse.

Chain severance can occur through multiple mechanisms—genetic (receptor damage), microenvironmental (signal loss), epigenetic (promoter silencing), or spatial (basement membrane breach)—all producing the same outcome (Table 4). This explains why cancers arising from different mechanisms (hereditary, environmental, sporadic) converge on the same phenotypic endpoints.

4 Engagement with Prior Work

Table 5 summarizes the relationship between existing cancer frameworks and chain severance.

The closest existing framework is TOFT [9]. It correctly identifies the organizational level (tissue, not cell) and the default proliferative state. Our framework provides the formalism TOFT has lacked: the escalation chain is the formal structure of tissue organization, C_n^{eff} is the measure of organizational integration, and chain severance is the formal description of organizational breakdown. TOFT's "default proliferative state" is our "necessary action space under cell-level C_n."

5 Conclusion

Counting organizational systems disrupted predicts cancer survival better than counting mutations. This result is robust across three independent datasets (73,593 patients), eight channel granularities, within individual cancer types, after adjusting for 275 clinical covariates, under three mutation-filtering stringencies, and against data-driven alternatives. Coupling channels are severed in escalation order: cell-intrinsic first, organism-level last.

The structural interpretation is chain severance: a cell that loses its coupling channels loses the effective context those channels provided and converges on cell-level optimization. The formal derivations appear in companion papers [5, 6, 7, 8].

The immediate question the empirical result raises is whether finer-grained distinctions within channels—specifically, the position of a mutation in the signaling graph (hub vs. leaf)—carry additional prognostic information beyond channel count alone. If so, the coupling-channel framework generates predictions about cancer subtype, metastatic tropism, and combination therapy design that are testable against these same datasets.

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Tables

Table 4: Mechanisms of chain severance. All four produce the same outcome: $\lambda_j \to 0$, context collapse to $C_n^{(0)}$, and cell-level optimization.

Mechanism	How $\lambda_j \to 0$	Example	Identified by
Genetic	Cell's receivers damaged	E-cadherin loss, p53/Rb mutation	SMT [10]
Microenvironmental	Signal no longer transmitted	Chronic inflammation, fibrosis, tissue damage	TOFT [9]
Epigenetic	Receiver genes silenced without sequence change	Methylation of connexin or cadherin promoters	—
Spatial	Cell leaves signaling field	Basement membrane breach, metastasis	—

Table 5: Relationship to existing cancer frameworks.

Framework	What it captures	Our reinterpretation
Hallmarks [4]	Universal capabilities across cancers	Not acquired but *revealed*: necessary action space under $C_n^{(0)}$
Atavistic theory [3]	Phylostratigraphic gene pattern	Convergent necessity, not historical reversion
Clonal evolution [16]	Darwinian selection dynamics	Provides the *attractor*: direction is determined by $C_n^{(0)}$, not random
Attractor landscape [17]	Cancer as abnormal attractors	"Abnormal" attractors are the *default* under cell-level C_n
Cancer as cheating [18]	Cooperation breakdown	Cheating is not a strategy but the necessary outcome of chain severance
Ecological process [19]	Tumor as ecosystem	Correct dynamics, but missing why they converge

Table 6: Complete gene-to-channel mapping used in all analyses. Genes are assigned based on organizational function served, not pathway membership alone. The full mapping (92 genes) and analysis code are provided in the supplementary repository.

Channel	**Genes**
DNA Damage Response	*ATM*, *ATR*, *BRCA1*, *BRCA2*, *PALB2*, *RAD51C*, *RAD51D*, *RAD51B*, *CHEK1*, *CHEK2*, *BAP1*, *BARD1*, *FANCA*, *FANCC*, *FANCD2*, *MLH1*, *MSH2*, *MSH6*, *PMS2*, *POLE*, *POLD1*
Cell Cycle / Apoptosis	*TP53*, *RB1*, *CDKN1A*, *CDKN1B*, *CDKN2A*, *CDKN2B*, *CDK4*, *CDK6*, *CCND1*, *CCNE1*, *MDM2*, *MDM4*, *MYC*, *MYCN*
PI3K / Growth Signaling	*PIK3CA*, *PIK3R1*, *PTEN*, *AKT1*, *AKT2*, *AKT3*, *MTOR*, *KRAS*, *NRAS*, *HRAS*, *BRAF*, *RAF1*, *MAP2K1*, *MAP2K2*, *MAP3K1*, *MAP3K13*, *ERBB2*, *ERBB3*, *EGFR*, *FGFR1*, *FGFR2*, *FGFR3*, *IGF1R*, *MET*, *NF1*, *NF2*, *TSC1*, *TSC2*, *STK11*, *ARID1A*
Endocrine / Hormone	*ESR1*, *ESR2*, *PGR*, *AR*, *FOXA1*, *GATA3*
Immune Surveillance	*B2M*, *HLA-A*, *HLA-B*, *HLA-C*, *JAK1*, *JAK2*, *STAT1*, *CD274*, *PDCD1LG2*, *CTLA4*
Tissue Architecture	*CDH1*, *CDH2*, *CTNNB1*, *APC*, *AXIN1*, *AXIN2*, *SMAD2*, *SMAD3*, *SMAD4*, *TGFBR1*, *TGFBR2*, *NOTCH1*, *NOTCH2*, *NOTCH3*, *NOTCH4*, *FBXW7*, *GJA1*, *GJB2*